Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2025

Course: B.Tech
Code: 101602

Subject: Construction Engineering & Management

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
*(iv) Question No. **1** is compulsory.*

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) The "S-Curve" in project management is used to represent
(i) Cost variation over time
(ii) Worker attendance
(iii) Equipment productivity
(iv) Soil settlement

(b) The purpose of a *Work Breakdown Structure (WBS)* is to
(i) Define the materials needed
(ii) Break the project into manageable activities
(iii) Estimate the total project cost
(iv) Allocate resources only for manpower

© Apna BEU

(c) In CPM, the total float of a critical activity is
(i) Zero
(ii) Positive
(iii) One
(iv) Negative

(d) PERT is best suited for projects that are
(i) Deterministic
(ii) Repetitive
(iii) Uncertain and research-oriented
(iv) Highly mechanized

(e) Equipment productivity depends primarily on
(i) Fuel cost
(ii) Operator skill, maintenance, and site conditions
(iii) Government regulations
(iv) Equipment color

(f) In PERT, the expected activity time is calculated as
(i) (To + Tp + Tm)/3
(ii) (Tm + Tp)/2
(iii) (To + 4Tm + Tp)/6
(iv) (To + 2Tm + Tp)/4

(g) Management is best defined as
(i) Planning and controlling only
(ii) Supervising machinery operation
(iii) Preparing construction drawings
(iv) Getting things done through others effectively and efficiently

(h) Which of the following is a *leading indicator* of safety performance on construction sites?
(i) Frequency rate of injuries
(ii) Number of safety training sessions conducted
(iii) Cost of compensation paid
(iv) None of these

(i) The agency responsible for financing and defining the objectives of a project is generally the
(i) Contractor
(ii) Consultant
(iii) Client
(iv) Supplier

(j) Which of the following is *not* a function of management?
(i) Planning
(ii) Controlling
(iii) Organizing
(iv) Manufacturing.

Apna BEU: Syllabus, PYQs & AI
All-in-One app for Syllabus, PYQs, Result, Attendance and more.
GET IT ON Google Play

Q.2	(a)	A project plan consisting of 14 activities A through N. The relationship amongst these activities is given below: (1) A is the first job to be performed, (2) B and C cab be done concurrently and must follow A, (3) D, E and F follow B, (4) G follows E, (5) H must succeed D but it can't start until E is complete, (6) I and J follows G, (7) F and J precede K, (8) H and I precede L, (9) M follows L and K, (10) The last operation N succeeds M and C. Construct the network diagram and identify critical path.	[7]
	(b)	Explain the unique features that distinguish construction projects from manufacturing projects.	[7]
Q.3	(a)	Describe the different types of construction projects and the agencies involved in their execution.	[7]
	(b)	Discuss the various phases of a construction project from conception to commissioning.	[7]
Q.4	(a)	Explain the role of the client and contractor during the pre-tender and pre-construction stages	[7]
	(b)	What are bar charts and Gantt charts? Discuss their advantages and limitations.	[7]
Q.5	(a)	A project has optimistic time = 6 days, pessimistic time = 18 days, and most likely time = 12 days. Calculate expected duration and standard deviation of the activity.	[7]
	(b)	Compare precast concrete and modular construction methods.	[7]
Q.6	(a)	Discuss the advantages of mechanized construction over conventional methods.	[7]
	(b)	Explain the importance of site layout and the factors affecting it.	[7]
Q.7	(a)	Explain the concept and significance of resource leveling and resource smoothing.	[7]
	(b)	Describe the use of histograms and S-curves in construction project management.	[7]
Q.8	(a)	What are the common causes of time and cost overruns in construction projects? Suggest corrective measures.	[7]
	(b)	Explain the role of inspection and statistical quality control in maintaining construction quality.	[7]
Q.9	(a)	Explain the concept of time-cost trade-off and the factors affecting it in project management	[7]
	(b)	Discuss the various types of construction contracts and their applications.	[7]

© Apna BEU

Apna BEU: Syllabus, PYQs & AI
All-in-One app for Syllabus, PYQs, Result, Attendance and more.
GET IT ON Google Play